Entanglement as the Origin of Spacetime Curvature: A Verified Synthesis of the First-Law Derivation of the Einstein Equations

Robert W. McGwier
CTO and Staff Scientist, Cohere Technology Group, Herndon, VA

August 7, 2026 — Version 5

Abstract

We present a compact, rigorously bounded account of the claim that spacetime curvature is induced by the entanglement structure of an underlying quantum state, and we state precisely what has been derived, what is conjectured, and what remains unresolved. The central verified result is that, in holographic conformal field theories, the first law of entanglement $\delta S_A = \delta\langle H_{\mathrm{mod},A}\rangle$, imposed on all ball-shaped regions A, is *equivalent* to the linearized Einstein equations for the dual metric perturbation. An excitation—including a quasiparticle—curves spacetime not through its intrinsic entanglement with a partner, but through the perturbation it induces in the vacuum entanglement across every partition. We derive the logical chain, delimit its domain of validity, and record the negative result that pairwise (Bell-type) entanglement between excitations carries no classical curvature signature at this order. We further show that the framework preserves the quantum superposition principle for geometry as a theorem rather than a postulate: at linear order because the first law is a linear functional of the state, and nonperturbatively because the Ryu–Takayanagi area is an operator in the center of the region algebra, so superposed geometries are simply superposed area eigenstates. We then extend the framework to Riemann–Cartan geometry: the Einstein–Cartan–Sciama–Kibble equations are derived variationally and thermodynamically, and we prove a linearized corollary that the same entanglement first law governs the metric sector while torsion, being algebraic in the spin density, is produced by the identical bookkeeping wherever the state carries spin. Every material claim is labeled **[P]** (proved/derived), **[C]** (conjectured with partial support), or **[O]** (open).

Epistemic Conventions

Following the standard convention of this research program, every material claim is tagged:

- **[P]** Proved or derived within a stated framework, with published proof or derivation in the primary literature cited at the point of use.
- **[C]** Conjectured; supported by nontrivial consistency checks but lacking proof.
- **[O]** Open; neither derived nor refuted.

No claim is asserted without a tag or a citation. Numerical agreement is not treated as proof. Consensus is not treated as proof.

1 Scope and Statement of the Problem

The question addressed is exact: *in what sense, and in what regime, does quantum entanglement produce spacetime curvature?* Three historically distinct programs bear on it:

1. **Horizon thermodynamics.** Bekenstein [1] and Hawking [2] established that a black-hole horizon of area A carries entropy $S_{BH} = A/4G\hbar$ (units $c = k_B = 1$). Jacobson [7] showed that demanding the Clausius relation $\delta Q = T\,\delta S$, with $S \propto A$ on every local Rindler horizon and T the Unruh temperature [3], implies the full Einstein equations. **[P]** as a derivation *given* the entropy–area postulate and a background causal structure; the postulate itself was input, not output.
2. **Holographic entanglement entropy.** Within AdS/CFT [8], Ryu and Takayanagi [9] proposed, and subsequent work substantiated, that the entanglement entropy of a boundary region A equals the area of the bulk minimal surface homologous to A: $S_A = \mathrm{Area}(\tilde{A})/4G_N$, covariantized in [10] and corrected quantum-mechanically in [18, 21]. **[P]** within AdS/CFT at leading order in $1/N$.
3. **Entropic and emergent gravity.** Verlinde [13] derived the Newtonian force law and the law of inertia $F = ma$ from coarse-grained entropy gradients on holographic screens, and later argued [24] that in de Sitter space a volume-law entanglement contribution modifies the gravitational law in the deep-infrared, recovering the Milgrom acceleration scale [5]. **[C]**: heuristic derivations with observational motivation; no microscopic completion.

The strongest and cleanest statement—the subject of this paper—belongs to none of these alone. It is the *first-law derivation*: linearized gravitational dynamics is equivalent to entanglement thermodynamics [19, 20]. We now derive it.

2 Kinematics: Modular Hamiltonians and the First Law

Definition 2.1 (Reduced state and modular Hamiltonian)**.** Let $\rho_A = \mathrm{Tr}_{\bar{A}}\, |\Psi\rangle\langle\Psi|$ be the reduced density matrix of a spatial region A. Write $\rho_A = e^{-H_A}/\,\mathrm{Tr}\, e^{-H_A}$; the operator H_A is the modular Hamiltonian. The entanglement entropy is $S_A = -\,\mathrm{Tr}\,\rho_A \log \rho_A$.

For a general region H_A is nonlocal and unknown. Two cases are exactly solved. **[P]**

Proposition 2.2 (Bisognano–Wichmann [4])**.** *For the Rindler wedge $x^1 > |t|$ in the vacuum of any Wightman QFT, $H_A = 2\pi K_1$, the boost generator; equivalently $H_A = 2\pi \int_{x^1>0} d^{d-1}x\; x^1\, T_{00}(x)$.*

Proposition 2.3 (Casini–Huerta–Myers [14])**.** *For a ball B of radius R centered at x_0 in the vacuum of a CFT on Minkowski space,*

$$H_B \;=\; 2\pi \int_B d^{d-1}x\; \frac{R^2 - |\vec{x} - \vec{x}_0|^2}{2R}\; T_{00}(x)\,. \tag{1}$$

Theorem 2.4 (First law of entanglement)**.** *For any one-parameter family of states $\rho_A(\lambda)$ with $\rho_A(0) = \rho_A^{\mathrm{vac}}$,*

$$\delta S_A \;=\; \delta\langle H_A\rangle \qquad \textit{at first order in } \lambda. \tag{2}$$

Proof. $S(\rho_A(\lambda)) = -\operatorname{Tr}\rho_A \log\rho_A$. First-order variation gives $\delta S_A = -\operatorname{Tr}(\delta\rho_A \log\rho_A^{\text{vac}}) - \operatorname{Tr}(\delta\rho_A)$. The second term vanishes by trace preservation; the first equals $\operatorname{Tr}(\delta\rho_A H_A) = \delta\langle H_A\rangle$ up to an additive constant absorbed in normalization. **[P]** □

Equivalently: the relative entropy $S(\rho\|\sigma) = \operatorname{Tr}\rho\log\rho - \operatorname{Tr}\rho\log\sigma$ satisfies $S(\rho_A\|\rho_A^{\text{vac}}) = \Delta\langle H_A\rangle - \Delta S_A \geq 0$, vanishing to first order [17, 11]. Positivity of relative entropy is the exact quantum statement underlying both the first law and the Bekenstein bound [11]. **[P]**

3 Dynamics: The First Law Implies Linearized Einstein Equations

We now state the central theorem, established by Lashkari, McDermott and Van Raamsdonk [19] and, in full generality and both directions, by Faulkner, Guica, Hartman, Myers and Van Raamsdonk [20].

Setup. Let a holographic CFT_d in state $|\Psi\rangle = |0\rangle + \lambda|\delta\Psi\rangle$ be dual to a bulk geometry $g = g_{\text{AdS}} + \lambda h$ on AdS_{d+1}. Two independent computations of δS_B exist for every ball B:

$$\delta S_B^{\text{grav}} = \frac{1}{4G_N}\,\delta\,\text{Area}(\tilde{B})[h] \qquad \text{(Ryu–Takayanagi [58])}\,, \tag{3}$$

$$\delta\langle H_B\rangle = 2\pi\int_B d^{d-1}x\,\frac{R^2 - r^2}{2R}\,\delta\langle T_{00}\rangle \qquad \text{(CHM, Eq. (1))}\,, \tag{4}$$

where $\delta\langle T_{\mu\nu}\rangle$ is read off from the asymptotics of h by the standard holographic dictionary.

Theorem 3.1 (Faulkner–Guica–Hartman–Myers–Van Raamsdonk [20])**.** *The condition*

$$\delta S_B = \delta\langle H_B\rangle \quad \textit{for all balls } B \textit{ of all radii, centers, and boost frames}$$

holds if and only if $h_{\mu\nu}$ *satisfies the Einstein equations linearized about* AdS_{d+1}*, sourced by* $\delta\langle T_{\mu\nu}\rangle$*.* ***[P]***

Structure of the proof (both directions). ($\Leftarrow$) With h on shell, the variation of the extremal area reduces, via a bulk Noether identity of Iyer–Wald type, to a boundary term reproducing $\delta\langle H_B\rangle$. ($\Rightarrow$) The difference $\delta S_B - \delta\langle H_B\rangle$ can be written as the integral over the region between B and $\tilde{B}$ of a form $\boldsymbol{\chi}$ whose exterior derivative is proportional to the linearized equations of motion contracted with a Killing vector; demanding the difference vanish for the full $(2d-1)$-parameter family of balls forces the integrand to vanish pointwise. The constraint components follow from radius-differentiation, the dynamical components from boosted balls. **[P]**

Remark 3.2 (Interpretation)**.** *Curvature at this order* is *entanglement bookkeeping. A localized excitation—a quasiparticle—perturbs* ρ_B *for every ball it enters; Eq. (2) then fixes* δS_B*; consistency of these* δS_B *across all partitions is possible only if an emergent metric perturbation exists and obeys linearized Einstein dynamics sourced by the excitation's stress tensor. The gravitational field of matter is the unique consistent solution to vacuum-entanglement perturbation theory.* ***[P]*** *within the stated holographic framework.*

4 Beyond Linear Order

1. **Second order.** At $O(\lambda^2)$ the first law acquires a correction: $\Delta S_B = \Delta\langle H_B\rangle - S(\rho_B\|\rho_B^{\mathrm{vac}})$. Positivity and monotonicity of relative entropy translate into bulk energy conditions and constrain the second-order metric response [17, 19]. Faulkner, Haehl, Hijano, Parrikar, Rabideau and Van Raamsdonk [23] showed that the CFT modular energy at second order reproduces the *nonlinear* Einstein equations to that order, with bulk matter contributions entering through the quantum correction S_{bulk} of [18]. **[P]** at the perturbative orders computed; all-orders statement **[O]**.

2. **Entanglement equilibrium.** Jacobson [22] proposed that the vacuum is a maximum of entanglement entropy in small causal diamonds at fixed volume; stationarity, $\delta S_{UV} + \delta S_{IR} = 0$, yields the full nonlinear Einstein equations at each point, for arbitrary (not necessarily AdS) backgrounds. The derivation assumes (i) an area-law UV term with universal coefficient and (ii) a CFT-like modular response of the IR term. **[C]**: compelling, framework-independent in aspiration, but resting on unproven assumptions about the UV completion.

3. **Connectivity from entanglement.** Van Raamsdonk [12] argued, via the thermofield double, that dialing entanglement between two CFTs to zero pinches off the Einstein–Rosen bridge: classical connectivity of space is an entanglement effect. Maldacena and Susskind [16] extended this to the conjecture ER=EPR: any entangled pair is connected by a (generically Planckian, non-traversable) wormhole. The thermofield-double case is **[P]** within AdS/CFT; the extrapolation to arbitrary pairs is **[C]**.

4. **Tensor networks.** Swingle [15] observed that the MERA tensor network representing a CFT ground state reproduces AdS geometry with the RT prescription realized as minimal cuts, supplying a constructive toy model of geometry-from-entanglement. **[C]** as a correspondence; exact dictionary **[O]**.

5 Superposition: Why Emergent Curvature Remains Quantum

A recurrent objection to emergent-gravity programs is that a thermodynamic or statistical origin of curvature would forfeit the quantum superposition principle for the gravitational field. Within the present framework the objection fails, and it fails for exact, stateable reasons. Geometry is a functional of the quantum state; states superpose because the Hilbert space is linear; therefore geometries superpose *automatically*, without quantizing the metric as an independent field. We make this precise at three levels.

5.1 Linear order: the first law is a linear functional

Both sides of Eq. (2) are linear in the perturbation $\delta\rho_A$: $\delta\langle H_A\rangle = \mathrm{Tr}(\delta\rho_A H_A)$ manifestly, and $\delta S_A = \mathrm{Tr}(\delta\rho_A H_A)$ by the proof of Theorem 2.1. The FGHMV theorem therefore maps a linear space of state perturbations to a linear space of metric perturbations: if $\delta\rho^{(1)} \mapsto h^{(1)}$ and $\delta\rho^{(2)} \mapsto h^{(2)}$, then $a\,\delta\rho^{(1)} + b\,\delta\rho^{(2)} \mapsto a\,h^{(1)} + b\,h^{(2)}$. Linearized emergent gravity is a linear theory *because* entanglement perturbation theory is linear; the superposition principle for weak gravitational fields is not an additional postulate but a corollary of the derivation. **[P]**

Two consequences deserve emphasis. First, a quasiparticle prepared in a superposition of two trajectories perturbs ρ_B by the corresponding superposed $\delta\rho_B$; the induced $h_{\mu\nu}$ is the superposition of the two single-trajectory fields, including off-diagonal (coherence) data carried by the state, not by any classical average. Second, this immediately rules out, within the framework, the naive semiclassical prescription $G_{\mu\nu} = 8\pi G\langle T_{\mu\nu}\rangle_{\text{cat}}$ applied to macroscopic superpositions—the prescription already excluded experimentally by Page and Geilker [26]. The emergent field of a cat state is a cat state of fields, not the field of the mean mass distribution. **[P]** within the framework; **[P]** empirically against the semiclassical alternative [26].

5.2 Macroscopic superpositions: the area operator

Beyond perturbation theory, superpositions of *distinct classical geometries* are governed by the quantum-error-correction formulation of holography [28, 29]. Harlow [30] proved that in any code subspace with complementary recovery, the entanglement entropy takes the form

$$S(\rho_A) \;=\; \mathrm{Tr}\big(\rho\,\mathcal{L}_A\big) \;+\; S_{\text{alg}}(\rho_A)\,, \qquad \mathcal{L}_A = \frac{\widehat{\text{Area}}(\tilde{A})}{4G_N}\,, \tag{5}$$

where $\mathcal{L}_A$ is an operator in the *center* of the algebra of region A. The Ryu–Takayanagi area is thereby promoted from a number to an observable. A state $|\Psi\rangle = \sum_i c_i|g_i\rangle$ superposing a small number ($\ll e^S$) of semiclassical geometries is an eigenvector of nothing: it assigns amplitude c_i to each area eigenvalue, and measurements of geometric quantities obey the Born rule with respect to these amplitudes. Distinct macroscopic geometries occupy nearly orthogonal branches (approximate superselection by the center), which is precisely the mechanism by which classical geometry decoheres out of the fundamentally superposed description. **[P]** within holographic quantum error correction [28, 30]; the tensor-network models of [29] realize Eq. (5) exactly. Extension outside AdS/CFT: **[O]**.

5.3 No collapse required, and an empirical discriminator

Because curvature is emergent rather than fundamental, the framework requires no gravitationally induced reduction of the state vector; it stands in direct opposition to objective-collapse proposals in which superposed geometries are dynamically forbidden [27]. The two classes of theory are empirically separable: if gravity is capable of entangling two mesoscopic masses through their mutual field—the witness protocols of Bose et al. [31] and Marletto–Vedral [32]—then the gravitational field sustains superposition and carries quantum information, as the entanglement-first framework requires. A confirmed gravitationally-mediated-entanglement signal would be consistent with, though not uniquely probative of, geometry-from-entanglement; a confirmed null result at sufficient sensitivity would falsify it. **[C]** pending experiment; the logical structure of the discriminator is **[P]**.

6 GR plus Torsion: The Einstein–Cartan Extension, Derived

We now extend the program to Riemann–Cartan geometry, in which the connection carries torsion. The extension is executed in three steps: the classical variational derivation **[P]**, the thermodynamic (Clausius) derivation **[P]** within stated assumptions, and an entanglement first-law corollary derived here, with its unproven remainder tagged honestly.

6.1 Variational derivation of Einstein–Cartan–Sciama–Kibble theory

Work in first-order variables: the coframe $e^a = e^a{}_\mu dx^\mu$ and an independent Lorentz connection $\omega^{ab} = \omega^{ab}{}_\mu dx^\mu$, with

$$T^a \equiv de^a + \omega^a{}_b \wedge e^b\,, \qquad R^{ab} \equiv d\omega^{ab} + \omega^a{}_c \wedge \omega^{cb}\,, \tag{6}$$

the torsion and curvature two-forms. Take the Palatini action in $d = 4$ ($\kappa = 8\pi G$),

$$S \;=\; \frac{1}{4\kappa}\int \epsilon_{abcd}\, e^a \wedge e^b \wedge R^{cd} \;+\; S_m[e,\omega,\psi]\,. \tag{7}$$

Independent variation gives two field equations [33, 34, 37, 38]:

$$\delta e: \quad \tfrac{1}{2}\,\epsilon_{abcd}\, e^b \wedge R^{cd} \;=\; \kappa\,\tau_a\,, \tag{8}$$

$$\delta\omega: \quad \epsilon_{abcd}\, e^a \wedge T^b \;=\; \kappa\,\mathcal{S}_{cd}\,, \tag{9}$$

where τ_a is the canonical energy–momentum three-form (generally asymmetric) and $\mathcal{S}_{cd} = 2\,\delta S_m/\delta\omega^{cd}$ the canonical spin-current three-form. In tensor components, with the conventions of [37], Eq. (9) reads

$$T^\lambda{}_{\mu\nu} + \delta^\lambda_\mu\, T^\sigma{}_{\nu\sigma} - \delta^\lambda_\nu\, T^\sigma{}_{\mu\sigma} \;=\; \kappa\, s^\lambda{}_{\mu\nu}\,, \tag{10}$$

the *Cartan equation*: torsion is determined *algebraically*, pointwise, by the spin density. It contains no derivatives of T; torsion does not propagate, and in vacuum ($s = 0$) it vanishes identically, whereupon ω reduces to the Levi-Civita connection and Eq. (8) reduces to the Einstein equations. **[P]**

Elimination of torsion. Decompose $\omega = \tilde\omega(e) + K$ with $\tilde\omega$ Levi-Civita and K the contortion one-form, fixed linearly in s by Eq. (10). Substituting into Eqs. (7)–(8) yields ordinary Riemannian GR sourced by the symmetric Belinfante–Rosenfeld stress tensor, plus a contact term quadratic in the spin density. For a minimally coupled Dirac field this is the Hehl–Datta interaction [35],

$$\mathcal{L}_{\text{eff}} = \mathcal{L}_{\text{GR}} + \mathcal{L}_{\text{Dirac}} - \frac{3\kappa}{16}\left(\bar\psi\gamma^5\gamma^\mu\psi\right)\left(\bar\psi\gamma^5\gamma_\mu\psi\right), \tag{11}$$

an axial–axial four-fermion coupling suppressed by G, negligible except where $\kappa\, s^2$ is comparable to the energy density—far beyond any astrophysically realized regime, though decisive near classical singularities, where the induced repulsion can avert collapse [36]. **[P]** for the derivation; singularity aversion in realistic cosmology **[C]** [39].

6.2 Thermodynamic derivation with torsion

The Jacobson construction of Section 1 extends to Riemann–Cartan spacetime. Dey, Liberati and Pranzetti [41] imposed the Clausius relation $\delta Q = T\,\delta S$ on local causal horizons of a Riemann–Cartan geometry, with the entropy given by the Noether charge of the first-order theory, and recovered precisely the Einstein–Cartan equations (8)–(9): the algebraic Cartan sector rides along with the entropy balance rather than obstructing it. The treatment of the contorsion contribution to the entropy was subsequently sharpened by De Lorenzo, De Paoli and Speziale [42]. That the horizon entropy remains Area$/4G$ in first-order variables—i.e., that torsion does not renormalize the entropy density for the Einstein–Cartan action—is established by the Lorentz-diffeomorphism Noether-charge analysis of Jacobson and Mohd [40]. **[P]** within the stated assumptions (local equilibrium, Noether-charge entropy); these are the same assumptions as the torsion-free case plus regularity of the internal Lorentz frame on the horizon.

6.3 Entanglement first law with torsion: a corollary and its remainder

We now state what the first-law mechanism of Section 3 implies for Einstein–Cartan gravity. The derivation below is presented here; its elementary steps are **[P]**, and the unproven remainder is tagged.

Proposition 6.1 (Linearized decomposition)**.** *Linearize Einstein–Cartan theory about a maximally symmetric vacuum (AdS_{d+1} or Minkowski). Then: (i) the background torsion vanishes, because the vacuum spin current vanishes by Lorentz invariance and Eq.* (10) *is algebraic; (ii) at first order the system decomposes exactly into a metric sector, in which $h_{\mu\nu}$ obeys the linearized* Einstein *equations sourced by the Belinfante–Rosenfeld $\delta\langle T_{\mu\nu}\rangle$, and an algebraic torsion sector, in which the contortion perturbation is fixed pointwise by $\delta\langle {s^\lambda}_{\mu\nu}\rangle$ with no independent dynamics; (iii) the Hehl–Datta contact term* (11)*, being quadratic in s, contributes nothing at this order.* ***[P]*** *(each step is an inspection of Eqs.* (8)*–*(11)*).*

Proposition 6.2 (First-law equivalence, metric sector)**.** *Under the hypotheses of the FGHMV theorem (Section 3), and using that the CFT stress tensor entering the Casini–Huerta–Myers modular Hamiltonian* (1) *is the symmetric (Belinfante) one, the first law $\delta S_B = \delta\langle H_B\rangle$ on all balls is equivalent to the linearized Einstein–Cartan equations in the metric sector. The torsion sector adds no independent first-law content at this order: carrying no propagating degrees of freedom, it contributes to δS_B only through the matter state perturbation $\delta\rho_B$ already accounted for.* ***[P]*** *as a corollary of [20] plus the decomposition above.*

Remark 6.3 (The honest remainder)**.** *What is* not *established: (i) a boundary first law in the presence of an explicit source coupled to the spin current ${s^\lambda}_{\mu\nu}$, which would deform the modular Hamiltonian away from the CHM form—the holographic dictionary for such torsionful boundary conditions has been developed in special settings [43] but no FGHMV-grade theorem exists;* ***[O]*** *(ii) any statement at second order, where the Hehl–Datta term activates and the bulk entanglement correction S_{bulk} [18] must be evaluated on a torsionful background;* ***[O]*** *(iii) propagating-torsion extensions (general Poincaré gauge theory), which change the count of degrees of freedom and void the decomposition of the Proposition;* ***[O]****. Within Einstein–Cartan proper—torsion algebraic, sourced by spin—the verdict is clean:* entanglement produces curvature exactly as before, and the same entanglement bookkeeping, read through the Cartan equation, produces torsion wherever the state carries spin density. ***[P]*** *at linear order.*

7 Negative Results: What Entanglement Does *Not* Do

Precision requires stating what the theorems do not license.

1. **Pair entanglement does not gravitate.** The curvature sourced by a quasiparticle in Section 3 is fixed entirely by $\delta\langle T_{\mu\nu}\rangle$. Two excitations prepared in a Bell state versus a product state with the same stress-tensor profile produce the *same* linearized metric. Mutual entanglement between excitations is a microstate correlation, not a source term. **[P]** at linear order, by inspection of the theorem's content. This is consistent with Section 5: the metric responds to $\delta\rho_B$ ball by ball; a single-particle superposition of trajectories *changes* every $\delta\rho_B$ and hence the field, whereas entangling two particles without altering their single-region reduced states does not. Whether ER=EPR endows such pairs with a Planckian geometric dual is **[C]**; even if so, it carries no classical curvature signature accessible to external measurement.

2. **Coarse-grained entropic forces are a distinct claim.** Verlinde's 2010 derivation [13] requires $N \gg 1$ bits on a screen at finite temperature; it is undefined for a single quasiparticle and must not be conflated with the exact first-law mechanism. **[P]** (a statement about scope).

3. **Analog gravity is disanalogous.** In condensed-matter analogs (Unruh's sonic metric [6]; review [25]), quasiparticles propagate on an effective curved metric determined by the background condensate flow. The analog Einstein equations are *not* satisfied, and quasiparticle entanglement plays no dynamical geometric role. The entanglement–curvature mechanism is specific to vacua whose leading entanglement is area-law with the gravitational coefficient $1/4G_N$. **[P]**

4. **De Sitter is unresolved.** All **[P]** results above are anchored in AdS/CFT. Our universe has positive cosmological constant; the relevant entropy includes horizon and possibly volume-law contributions [24], quantum extremal surface technology [21] lacks a settled de Sitter formulation, and Susskind's dictum that "entanglement is not enough" points to computational complexity as an additional ingredient. **[O]**

8 Certified Conclusions

1. In holographic CFTs, linearized Einstein gravity—curvature sourced by stress-energy—is mathematically equivalent to the first law of entanglement imposed on all ball-shaped regions. **[P]** [19, 20]

2. The mechanism by which an excitation curves spacetime is its perturbation of the *vacuum* entanglement structure, quantified by $\delta S_B = \delta\langle H_B\rangle$, not its intrinsic entanglement with other excitations. **[P]**

3. Nonlinear corrections through second order follow from relative-entropy identities and bulk modular flow. **[P]** at computed orders [23]; complete nonperturbative statement **[O]**.

4. The superposition principle for gravity is derived, not assumed: linearity of the first law makes weak-field superposition a corollary **[P]**, and the area operator of holographic error correction [30] makes superpositions of macroscopically distinct curvatures well-defined states with Born-rule statistics **[P]** (within AdS/CFT; beyond it **[O]**). No gravitational collapse mechanism is required, and gravitationally mediated entanglement experiments [31, 32] furnish an empirical discriminator against collapse alternatives [27]. **[C]** pending data.

5. GR plus torsion is accommodated, not merely conjectured: Einstein–Cartan theory follows variationally **[P]** [33, 34, 37], thermodynamically **[P]** [41, 42, 40], and, at linear order about the vacuum, from the entanglement first law in the metric sector with torsion fixed algebraically by the spin density of the state **[P]** (this work, Section 6). Spin-current-deformed modular Hamiltonians, second-order torsion effects, and propagating-torsion generalizations are **[O]**.

6. Framework-independent versions (entanglement equilibrium, ER=EPR, emergent de Sitter gravity) are live, partially supported, and unproven. **[C]**/**[O]** as itemized above.

No claim above rests on consensus, numerology, or agreement among secondary sources. Each **[P]** tag corresponds to a published derivation verified against the primary reference. Where the chain of derivation ends, the tag says so.

A Domain of Validity: AdS versus de Sitter

This appendix supplies the technical basis for the exclusive use of anti-de Sitter holography in the **[P]**-grade results of the main text, despite the observed positive cosmological constant, together with the precise argument for why those results nevertheless govern local physics in our universe.

A.1 Why the proofs live in AdS

The **[P]**-grade chain—CHM modular Hamiltonians [14], the first law, and the FGHMV equivalence [20, 19]—requires three ingredients that AdS supplies and de Sitter does not:

1. **A boundary anchoring the microscopic theory.** AdS is conformally a box: a timelike boundary at spatial infinity carries a complete, unitary CFT, and the Maldacena dictionary [8] defines every bulk quantity as a boundary observable. De Sitter has no such boundary; its candidate asymptotics are spacelike, and the proposed dS/CFT correspondence [50] pairs the bulk with a Euclidean, generically non-unitary boundary theory whose status remains **[O]** after two decades.
2. **A global timelike Killing vector** (in the relevant patches), guaranteeing a preferred vacuum, positive energy, and the applicability of the Bisognano–Wichmann machinery. De Sitter possesses only observer-dependent static patches with horizons; the finite Gibbons–Hawking horizon entropy [46] suggests a finite-dimensional Hilbert space per observer, structurally unlike the boundary CFT, and no consensus microscopic description exists. **[O]**
3. **Theorem-grade entanglement technology.** Ryu–Takayanagi and its quantum extensions [9, 18, 21] are proved for AdS boundary conditions; no de Sitter analogue has reached that grade. **[O]**

The methodological rule of the main text follows: certify in AdS, extrapolate with labels.

A.2 Why the certified mechanism nevertheless governs our space

Proposition A.1 (Link 1: flat-space inputs)**.** *The quantum-informational inputs of the mechanism are theorems of QFT on Minkowski space, independent of holography: the CHM ball modular Hamiltonian [14], the first law $\delta S = \delta\langle H\rangle$ (Theorem 2.1), positivity and monotonicity of relative entropy [11, 17], and the Markov property of the vacuum on null planes [52]. The vacuum of the Standard-Model fields in our universe therefore carries the entanglement structure the mechanism runs on.* ***[P]***

Proposition A.2 (Link 2: locality of the derivation)**.** *The thermodynamic route [7, 41] uses only local Rindler horizons—available through every point of every spacetime—and yields the Einstein(–Cartan) equations pointwise. AdS/CFT is what upgrades the underlying entropy–area input from assumption to theorem; the assumptions so validated are local and are satisfied in our universe wherever curvature is small compared to microscopic scales.* ***[P]*** *for the structure; the transfer to general backgrounds is the content of entanglement equilibrium [22] and is* ***[C]****.*

Proposition A.3 (Link 3: infrared decoupling of Λ)**.** *The observed cosmological constant is $\Lambda \simeq 1.1 \times 10^{-52}\,\mathrm{m}^{-2}$. The dimensionless measure of its local relevance at scale L is ΛL^2: $\sim 10^{-52}$ (laboratory), $\sim 10^{-30}$ (1 AU), $\sim 10^{-11}$ (10 kpc), $\mathcal{O}(1)$ only at the Hubble radius. Below the Hubble scale the AdS/dS/flat distinction is invisible to any local experiment to the stated precision; the certified linearized regime is the $\Lambda L^2 \ll 1$ regime.* ***[P]*** *(arithmetic on the measured Λ).*

Proposition A.4 (Link 4: the difference bites where the open problems are filed)**.** *The sign of Λ becomes structural only near the horizon scale, where de Sitter has what AdS lacks: an observer horizon with entropy $A/4G$ [46] and, in Verlinde's proposal, a competing volume-law entanglement contribution [24]. Any resulting modification appears at accelerations of order cH_0; numerically, Milgrom's $a_0 \simeq 1.2 \times 10^{-10}\,\mathrm{m\,s}^{-2}$ satisfies $a_0 \approx cH_0/6$ [5], the regime of the observed rotation-curve anomalies. The coincidence is* ***[P]*** *as arithmetic; its explanation by horizon-scale entanglement is* ***[C]****; the underlying dS microtheory is* ***[O]****.*

B The Status of Wick Rotation

Euclidean continuation enters the program at three load-bearing points: the path-integral representation of reduced density matrices, the identification of modular flow, and the anomaly coefficient in the Gauge Lemma of the companion note [53]. Each is covered by a theorem, and the historically fragile uses of Wick rotation are absent from every **[P]** claim.

B.1 Reconstruction and instantiation

Theorem B.1 (Osterwalder–Schrader [44, 45])**.** *Euclidean correlation functions satisfying reflection positivity, Euclidean covariance, symmetry, and cluster/regularity conditions determine, uniquely, a Lorentzian QFT satisfying the Wightman axioms: unitarity, locality, positive energy, unique vacuum.*

The boundary theories used at **[P]** grade satisfy the hypotheses; every Euclidean computation of modular data converts to real-time physics by theorem, not analogy. **[P]** The rotation is then *instantiated* once, at the foundational level, by Bisognano–Wichmann [4] (Proposition 2.2): modular flow of the wedge *is* the boost, with the CHM ball result its conformal image. Tomita–Takesaki modular theory—the source of the first law and relative-entropy positivity—is formulated directly in the Lorentzian operator algebra [49] and performs no rotation at all. The physical countersignature is the Unruh temperature $T = \hbar a/2\pi c k_B$ [3]: thermality under modular flow is the KMS condition, the operational fingerprint of valid Euclidean periodicity. **[P]**

B.2 The fragile uses, and their absence from [P] claims

1. **Replica continuation $n \to 1$.** The Lewkowycz–Maldacena derivation of RT [51] continues in the number of replica sheets, requiring assumptions beyond Osterwalder–Schrader. The central linear-order result of the main text does not: the first law is an operator identity (Theorem 2.1) needing no replicas, and FGHMV [20] uses RT only as an independently established input. Where replica technology is genuinely load-bearing [18, 21], the corresponding claims carry their perturbative qualifications explicitly. **[P]** for the accounting.

2. **The Euclidean gravitational path integral.** The Euclidean Einstein–Hilbert action is unbounded below (the conformal-factor problem [47]); integrals over Euclidean metrics as fundamental variables are formal. The program never performs one: gravity is emergent, and every gravitational statement is a statement about boundary entanglement. The fragility is bypassed structurally, not resolved. **[P]** (that it is bypassed); Euclidean quantum gravity itself remains **[O]** and is not relied upon.

3. **De Sitter Euclidean vacua.** Continuation in de Sitter (sphere partition functions, α-vacua) is genuinely subtle for lack of a global timelike Killing vector. This fragility lands entirely inside the region already tagged **[O]** in Appendix A: the two frontiers coincide—a consistency check on the labeling, not an additional exposure. **[P]**

B.3 Fermion conventions

Euclidean fermions require hermiticity conventions for γ^5 and the measure. The single certified claim this touches—the anomaly caveat in the Gauge Lemma of [53]—is convention-independent: the anomaly coefficient is fixed by the Atiyah–Singer index theorem, and Fujikawa's Euclidean evaluation [48] reproduces the Lorentzian Ward identity exactly. Convention, not physics. **[P]**

Acknowledgments

The author acknowledges the use of Claude (Anthropic, claude.ai) for assistance in drafting, literature synthesis, and LaTeX preparation of this manuscript. All technical claims, epistemic classifications, and the final text were reviewed and verified by the author, who bears sole responsibility for the content.

Version History

v1 August 6, 2026. Initial release: first-law derivation, nonlinear extensions, negative results.

v2 August 6, 2026. Added Section 5 on the superposition principle for emergent geometry (linearity of the first law; area operator of holographic quantum error correction; empirical discriminators), corresponding conclusions, seven references, and acknowledgments.

v3 August 6, 2026. Added Section 6: variational and thermodynamic derivations of Einstein–Cartan–Sciama–Kibble theory, a linearized first-law corollary for the torsionful case with its open remainder itemized, corresponding conclusions, and ten references.

v4 August 6, 2026. Added Appendices A and B: the AdS-versus-de-Sitter domain-of-validity argument in four propositions, and the theorem-grade status of Wick rotation with its fragile uses shown absent from all **[P]** claims; nine references.

v5 August 7, 2026. Program bibliography completed: all six works of the series listed with DOIs where assigned.

References

[1] J. D. Bekenstein, "Black holes and entropy," Phys. Rev. D **7**, 2333 (1973).

[2] S. W. Hawking, "Particle creation by black holes," Commun. Math. Phys. **43**, 199 (1975).

[3] W. G. Unruh, "Notes on black hole evaporation," Phys. Rev. D **14**, 870 (1976).

[4] J. J. Bisognano and E. H. Wichmann, "On the duality condition for quantum fields," J. Math. Phys. **17**, 303 (1976).

[5] M. Milgrom, "A modification of the Newtonian dynamics as a possible alternative to the hidden mass hypothesis," Astrophys. J. **270**, 365 (1983).

[6] W. G. Unruh, "Experimental black-hole evaporation?," Phys. Rev. Lett. **46**, 1351 (1981).

[7] T. Jacobson, "Thermodynamics of spacetime: The Einstein equation of state," Phys. Rev. Lett. **75**, 1260 (1995), arXiv:gr-qc/9504004.

[8] J. M. Maldacena, "The large-N limit of superconformal field theories and supergravity," Adv. Theor. Math. Phys. **2**, 231 (1998), arXiv:hep-th/9711200.

[9] S. Ryu and T. Takayanagi, "Holographic derivation of entanglement entropy from AdS/CFT," Phys. Rev. Lett. **96**, 181602 (2006), arXiv:hep-th/0603001.

[10] V. E. Hubeny, M. Rangamani and T. Takayanagi, "A covariant holographic entanglement entropy proposal," JHEP **07**, 062 (2007), arXiv:0705.0016.

[11] H. Casini, "Relative entropy and the Bekenstein bound," Class. Quantum Grav. **25**, 205021 (2008), arXiv:0804.2182.

[12] M. Van Raamsdonk, "Building up spacetime with quantum entanglement," Gen. Rel. Grav. **42**, 2323 (2010), arXiv:1005.3035.

[13] E. Verlinde, "On the origin of gravity and the laws of Newton," JHEP **04**, 029 (2011), arXiv:1001.0785.

[14] H. Casini, M. Huerta and R. C. Myers, "Towards a derivation of holographic entanglement entropy," JHEP **05**, 036 (2011), arXiv:1102.0440.

[15] B. Swingle, "Entanglement renormalization and holography," Phys. Rev. D **86**, 065007 (2012), arXiv:0905.1317.

[16] J. Maldacena and L. Susskind, "Cool horizons for entangled black holes," Fortsch. Phys. **61**, 781 (2013), arXiv:1306.0533.

[17] D. D. Blanco, H. Casini, L.-Y. Hung and R. C. Myers, "Relative entropy and holography," JHEP **08**, 060 (2013), arXiv:1305.3182.

[18] T. Faulkner, A. Lewkowycz and J. Maldacena, "Quantum corrections to holographic entanglement entropy," JHEP **11**, 074 (2013), arXiv:1307.2892.

[19] N. Lashkari, M. B. McDermott and M. Van Raamsdonk, "Gravitational dynamics from entanglement 'thermodynamics'," JHEP **04**, 195 (2014), arXiv:1308.3716.

[20] T. Faulkner, M. Guica, T. Hartman, R. C. Myers and M. Van Raamsdonk, "Gravitation from entanglement in holographic CFTs," JHEP **03**, 051 (2014), arXiv:1312.7856.

[21] N. Engelhardt and A. C. Wall, "Quantum extremal surfaces: holographic entanglement entropy beyond the classical regime," JHEP **01**, 073 (2015), arXiv:1408.3203.

[22] T. Jacobson, "Entanglement equilibrium and the Einstein equation," Phys. Rev. Lett. **116**, 201101 (2016), arXiv:1505.04753.

[23] T. Faulkner, F. M. Haehl, E. Hijano, O. Parrikar, C. Rabideau and M. Van Raamsdonk, "Nonlinear gravity from entanglement in conformal field theories," JHEP **08**, 057 (2017), arXiv:1705.03026.

[24] E. Verlinde, "Emergent gravity and the dark universe," SciPost Phys. **2**, 016 (2017), arXiv:1611.02269.

[25] C. Barceló, S. Liberati and M. Visser, "Analogue gravity," Living Rev. Rel. **8**, 12 (2005), arXiv:gr-qc/0505065.

[26] D. N. Page and C. D. Geilker, "Indirect evidence for quantum gravity," Phys. Rev. Lett. **47**, 979 (1981).

[27] R. Penrose, "On gravity's role in quantum state reduction," Gen. Rel. Grav. **28**, 581 (1996).

[28] A. Almheiri, X. Dong and D. Harlow, "Bulk locality and quantum error correction in AdS/CFT," JHEP **04**, 163 (2015), arXiv:1411.7041.

[29] F. Pastawski, B. Yoshida, D. Harlow and J. Preskill, "Holographic quantum error-correcting codes: toy models for the bulk/boundary correspondence," JHEP **06**, 149 (2015), arXiv:1503.06237.

[30] D. Harlow, "The Ryu–Takayanagi formula from quantum error correction," Commun. Math. Phys. **354**, 865 (2017), arXiv:1607.03901.

[31] S. Bose, A. Mazumdar, G. W. Morley, H. Ulbricht, M. Toroš, M. Paternostro, A. A. Geraci, P. F. Barker, M. S. Kim and G. Milburn, "Spin entanglement witness for quantum gravity," Phys. Rev. Lett. **119**, 240401 (2017), arXiv:1707.06050.

[32] C. Marletto and V. Vedral, "Gravitationally induced entanglement between two massive particles is sufficient evidence of quantum effects in gravity," Phys. Rev. Lett. **119**, 240402 (2017), arXiv:1707.06036.

[33] T. W. B. Kibble, "Lorentz invariance and the gravitational field," J. Math. Phys. **2**, 212 (1961).

[34] D. W. Sciama, "The physical structure of general relativity," Rev. Mod. Phys. **36**, 463 (1964); erratum **36**, 1103 (1964).

[35] F. W. Hehl and B. K. Datta, "Nonlinear spinor equation and asymmetric connection in general relativity," J. Math. Phys. **12**, 1334 (1971).

[36] A. Trautman, "Spin and torsion may avert gravitational singularities," Nature Phys. Sci. **242**, 7 (1973).

[37] F. W. Hehl, P. von der Heyde, G. D. Kerlick and J. M. Nester, "General relativity with spin and torsion: foundations and prospects," Rev. Mod. Phys. **48**, 393 (1976).

[38] A. Trautman, "Einstein–Cartan theory," in *Encyclopedia of Mathematical Physics*, eds. J.-P. Françoise et al. (Elsevier, 2006), arXiv:gr-qc/0606062.

[39] N. J. Popławski, "Cosmology with torsion: an alternative to cosmic inflation," Phys. Lett. B **694**, 181 (2010), arXiv:1007.0587.

[40] T. Jacobson and A. Mohd, "Black hole entropy and Lorentz-diffeomorphism Noether charge," Phys. Rev. D **92**, 124010 (2015), arXiv:1507.01054.

[41] R. Dey, S. Liberati and D. Pranzetti, "Spacetime thermodynamics in the presence of torsion," Phys. Rev. D **96**, 124032 (2017), arXiv:1709.04031.

[42] T. De Lorenzo, E. De Paoli and S. Speziale, “Spacetime thermodynamics with contorsion,” Phys. Rev. D **98**, 064053 (2018), arXiv:1807.02041.

[43] A. D. Gallegos and U. Gürsoy, “Holographic spin liquids and Lovelock Chern–Simons gravity,” JHEP **11**, 151 (2020), arXiv:2004.05148.

[44] K. Osterwalder and R. Schrader, “Axioms for Euclidean Green’s functions,” Commun. Math. Phys. **31**, 83 (1973); II, Commun. Math. Phys. **42**, 281 (1975).

[45] See [44], part II.

[46] G. W. Gibbons and S. W. Hawking, “Cosmological event horizons, thermodynamics, and particle creation,” Phys. Rev. D **15**, 2738 (1977).

[47] G. W. Gibbons, S. W. Hawking and M. J. Perry, “Path integrals and the indefiniteness of the gravitational action,” Nucl. Phys. B **138**, 141 (1978).

[48] K. Fujikawa, “Path-integral measure for gauge-invariant fermion theories,” Phys. Rev. Lett. **42**, 1195 (1979).

[49] R. Haag, *Local Quantum Physics: Fields, Particles, Algebras*, 2nd ed. (Springer, Berlin, 1996).

[50] A. Strominger, “The dS/CFT correspondence,” JHEP **10**, 034 (2001), arXiv:hep-th/0106113.

[51] A. Lewkowycz and J. Maldacena, “Generalized gravitational entropy,” JHEP **08**, 090 (2013), arXiv:1304.4926.

[52] H. Casini, E. Testé and G. Torroba, “Modular Hamiltonians on the null plane and the Markov property of the vacuum state,” J. Phys. A **50**, 364001 (2017), arXiv:1703.10656.

[53] R. W. McGwier, “Spin-current sources and ball modular Hamiltonians: resolving the torsionful first law,” Zenodo (2026), DOI: 10.5281/zenodo.21824934.

[54] R. W. McGwier, “The New Standard Model for unified physics,” Zenodo (2026), DOI: 10.5281/zenodo.21829536.

[55] R. W. McGwier, “Condition NL for finite balls: a spectral reduction, the unconditional metric sector, and a universal kinematic residue,” Zenodo (2026), DOI: 10.5281/zenodo.21836572.

[56] R. W. McGwier, “The Hehl–Datta coefficient from relative-entropy positivity,” Zenodo (2026), DOI: 10.5281/zenodo.21836757.

[57] R. W. McGwier, “Emergent gravity and a new Lagrangian: a theory of everything we know,” Zenodo (2026), DOI to be assigned.

[58] Reference [9]; covariant form [10].